Energy & Buildings 366 (2026) 117700

Contents lists available at ScienceDirect

Energy & Buildings

journal homepage: www.elsevier.com/locate/enbuild

Mist evaporative cooling system with natural ventilation in a semi-outdoor environment: Theoretical models, experiments, and thermal simulations

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ARTICLE INFO

Keywords:
Evaporative cooling
Mist
Natural ventilation
Semi-outdoor
Thermodynamic modeling
Energyplus

ABSTRACT

Semi-outdoor environments in climates with a hot and dry season can achieve comfortable air temperature using mist evaporative cooling systems (MECS). However, assessing their effect using building thermal simulation programs requires a physical model of the MECS operating in combination with natural ventilation in semi-outdoor environments (MECS&NV) that can be implemented in such programs. This research presents, step by step, the development of a thermodynamic model for the MECS&NV process based on psychrometric theory, which can be implemented in EnergyPlus using its Python plug-in system. The developed model was validated against experimental results obtained in a semi-outdoor cafeteria. For the cafeteria air temperature, the mean absolute error is 0.4 °C and the mean bias error is 0.1 °C. The cooling effect of the MECS on the cafeteria air temperature compared with the value it would have without the MECS has a maximum reduction of 2.9 °C and an average of 1.9 °C. The developed model can be used to perform future simulations and support decision-making regarding sizing and implementation of this type of evaporative cooling system in semi-outdoor spaces for low-energy buildings.

1. Introduction

Evaporative cooling (EC) to increase people thermal comfort in indoor, semi-outdoor, and outdoor environments has been used for many centuries. Examples of its early applications can be found in the Alhambra, in Granada (Spain), built between the 12th and 14th centuries [1]. EC is more effective in climates having a hot season and low or intermediate air relative humidity. Direct evaporative cooling (DEC) is achieved when water is evaporated in the air, producing a decrease in air temperature and an increase in humidity. The air temperature is reduced because water phase change requires energy to happen, and this is taken from the air. Indirect evaporative cooling (IEC) is used when the increase in air humidity is not convenient. In this process, the air that is cooled by water evaporation passes through a heat exchanger with the air to be introduced into the space to be cooled. Due to the additional process, the IEC is less efficient than the DEC.

EC is considered a low-energy system as it consumes less energy than conventional cooling air conditioning systems. Commercial DEC and IEC units have been used to cool spaces with large volumes like theaters and auditoriums, which, when cooled with conventional cooling air conditioning systems, would require the use of large rooftop package equipment, ducts, and air terminals, with the inherent large energy consumption.

Due to the concern of climate change, there is a growing interest in the scientific community to develop EC systems for thermal comfort. Cuce and Riffat [2] presented a review of the subject, covering direct, indirect, and combined systems. Recently, advances have been made in EC systems with controlled airflow for indoor spaces without natural ventilation [3–5].

In the present research, the interest is on DEC used in conjunction with natural ventilation. The first studies found in the literature that uses DEC to cool the air that enters the building by natural ventilation employed wind towers (WT) or similar structures where water evaporation occurs. This type of cooling is named by some authors as passive downdraught evaporative cooling (PDEC). WT is a tall structure attached to a building in order to bring air into it. In Iran, some traditional WT incorporated DEC by allowing groundwater to seep through the inside of the basement walls [6]. The idea to incorporate DEC using water sprays on the top of a WT was proposed by Bahadori [7] in 1985. Since then, some authors have made theoretical models to simulate the process [7–10]. The first study of PDEC using computational fluid dynamics (CFD) for droplet evaporation was presented by Bowman et al. [11]. They

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https://doi.org/10.1016/j.enbuild.2026.117700
Received 3 March 2026; Received in revised form 28 April 2026; Accepted 25 May 2026
Available online 28 May 2026
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developed their one code. Subsequently, CFD studies mainly using commercial programs such as Fluent [6,12–14] and OpenFoam [15] have been reported, since these programs had already incorporated models for water evaporation. Kalantar [6] found that DEC by water sprays in a WT of 10 m height is able to reduce the air temperature up to 15 °C. Saffari and Hosseinnia [15] studied the DEC produced by wet curtains hung inside a WT. Wet curtains of 10 m in height decreased 10 °C the air temperature. Foroozesh et al. [13] analyzed the effect inside a room produced by a one-side WT with a water spray and a baffle in its interior. They found that a baffle inclined 60° provided more uniform velocity and temperature fields. Moghtader Gilvaei et al. [14] analyzed the effect of a WT that introduces air to a house via an earth-to-air heat exchanger and wet panels.

There are also experimental studies of PDEC reported in the literature [8,16,17]. Pearlmutter et al. [16] analyzed the performance of two cooling tower prototypes with DEC systems in Negev Highlands (Israel) with an arid climate. The prototype one, 3.5 m high, was installed in a small semi-enclosed courtyard. Different air circulation mechanisms and types of water supply were tested. Based on the results, they constructed the second prototype with a 10 m high WT with DEC assisted by a propeller fan installed in a 500 m^2 glazed courtyard. This system achieved an air temperature reduction of the air in the cooling tower up to 15 °C with a water consumption rate of 2 m^3/day. Belarbi et al. [8] constructed a building in Catania (Italy), with water sprays at the top of a 10 m height WT. They made measurements during the summer to test their theoretical model. Rabani et al. [17] studied a test room with a volume of less than 17 m^3 and a Trombe wall in combination with a solar chimney and a water spraying system, built in Yazd (Iran), with a desert climate. The indoor air temperature decreased up to 8 °C compared to the outdoor air, provided an average air change per hour (ACH) of 8, with a water mass flow rate of 10 l/h and a droplet diameter of 30 μm.

The first study of DEC in an outdoor environment was reported by Nishimura et al. [18] in 1998. They made measurements of wind velocity, air temperature, and humidity around an artificial water facility in Osaka (Japan) and confirmed the temperature reduction on the leeward side. Wong and Chong [19] performed measurements in a semi-outdoor space in Singapore with a hot and humid climate to evaluate the effect of the use of a commercial misting fan. They found that the misting fan significantly reduced the temperature and thermal sensation votes. Nevertheless, they pointed out the concern about the increase of bacterial and fungal pollutants under the misting fan system. Wei and He [20] presented a numerical model to predict the evaporative cooling effect of wet urban surfaces. They validated their model with experimental results, and by integrating the model into a 3D-CAD-based thermal simulation tool, they analyzed the DEC effect in two study cases. Ulpiani et al. [21] conducted measurements and surveys of the outdoor thermal comfort improvement with an overhead mist cooling system. They found that the temperature and the Universal Thermal Climate Index (UTCI) were reduced by 8.2 °C and 7.9 °C respectively, with an increase of 7% in relative humidity in the air just down the mist. The system proved to work best with a dominant and steady light breeze (1-2 m/s), in highly irradiated sites and suspended at 1.2-1.5 m above the average height of users. Heidari et al. [22] performed CFD simulations using ANSYS Fluent to analyze the effect on a residential building's thermal conditions of a horizontal windcatcher and an evaporative cooling system formed by a channel with a water falling film. Kachmar et al. [23] reported a numerical study using the ENVI-met urban microclimate model to evaluate the effectiveness of an outdoor windcacher and mist cooling system on an urban canyon in Doha, Qatar. The results indicate that the system reduces the air temperature in the sidewalk close to the system by more than 2 °C and the air temperature in the sidewalk far from the system by less than 1 °C; and increases the air velocity in the sidewalk close to the system by more than 0.5 m/s and in the sidewalk far from the system by less than 0.03 m/s.

There are studies that addressed the heat and mass processes between a water spray and the ambient air [24–27]. Sureshkumar et al. [24] performed experiments in a wind tunnel, varying the air dry bulb temperature (35 to 47 °C) and the relative humidity (10 to 60%). Two configurations were tested (parallel and counter flow), each with four nozzle sizes (hole diameter of 3, 4, 5, and 5.5 mm); three water pressures (1, 2, and 3 bar(g)) and three air velocities (1, 2 and 3 m/s). They found that for a specific water flow rate, a smaller nozzle at higher pressure produced more cooling than a larger nozzle at lower pressure. In a subsequent study, Sureshkumar et al. [25] proposed a simulation model for the two-way coupling of heat and mass transfer between the droplets and air. They performed two-dimensional CFD simulations to reproduce the experimental conditions. They found the best agreement between experiments and simulations for the parallel flow configuration. Montazeri et al. [26] used the experiments of [24] to validate their CFD simulations. Using CFD simulations considering the wind-tunnel test section, they carried out a parametric analysis of the DEC provided by a water spray system with a hollow-cone nozzle configuration. They used as evaluation parameters the sensible cooling and the UTCI, with respect to the inlet conditions, as functions of the nozzle distance. They found that the cooling performance is enhanced as the drop mean diameter decreases and drop-size distribution increases. The authors explained that they used the UTCI because the water spray systems are generally applied in outdoor or semi-outdoor environments. Younes et al. [27] performed wind tunnel experiments and CFD simulations using ANSYS Fluent to analyze the effect of incorporating fluctuations into the flow produced by a fan used in combination with mist cooling. They found that lower frequencies and higher amplitudes led to higher evaporation rates.

EnergyPlus is a program for whole-building thermal and energy simulations. It has the capability to simulate mechanical ventilation DEC systems (Zone HVAC:Evaporative Cooler Unit) and natural ventilation PDEC systems (defined as an Airflow object ZoneCoolTower:shower) [28]. Studies simulating a DEC unit in EnergyPlus have been reported in [29,30] and using Design Builder as an interface of EnergyPlus in [31,32]. A study simulating a PDEC in EnergyPlus using their own model was presented in [33].

The DEC of interest is a water spray evaporative cooling system, with small water droplets that generate a mist. This system is named the mist evaporative cooling system (MECS) [34,35]. The EnergyPlus DEC systems models can not simulate a MECS combined with natural ventilation (MECS&NV) in a semi-outdoor environment and to the best of the authors' knowledge, there are no models reported in the literature of a DEC system used directly inside a naturally ventilated environment to be used in EnergyPlus simulations.

Therefore, the aim of the research presented in this paper is the development of a model to obtain from EnergyPlus simulations the thermal conditions of a semi-outdoor environment that has a MECS&NV. The model is validated by comparing simulation results to experimental measurements.

The paper is structured as follows. Section 2 describes the studied system. The methodology of this research is presented in Section 3, which includes theoretical models, experiments, and thermal simulations. The results are displayed in Section 4. Discussion and limitations are presented in Section 5 and the conclusions are remarked in Section 6.

2. The studied system

The studied system is a mist evaporative cooling system (MECS) combined with natural ventilation (MECS&NV). The MECS uses a misting line with an array of nozzles. The nozzles have small holes in order to generate droplets of water with a diameter in the order of microns, that is, to generate a mist. This enhances the evaporation process with respect to a system with larger droplets. The only mechanical equipment needed is the water compressor to supply the water to the nozzles. This type of system is easy to install; the connection used between the water compressor and the nozzles is a small hose. The compressor and water tank could be located on the roof or basement of the building without

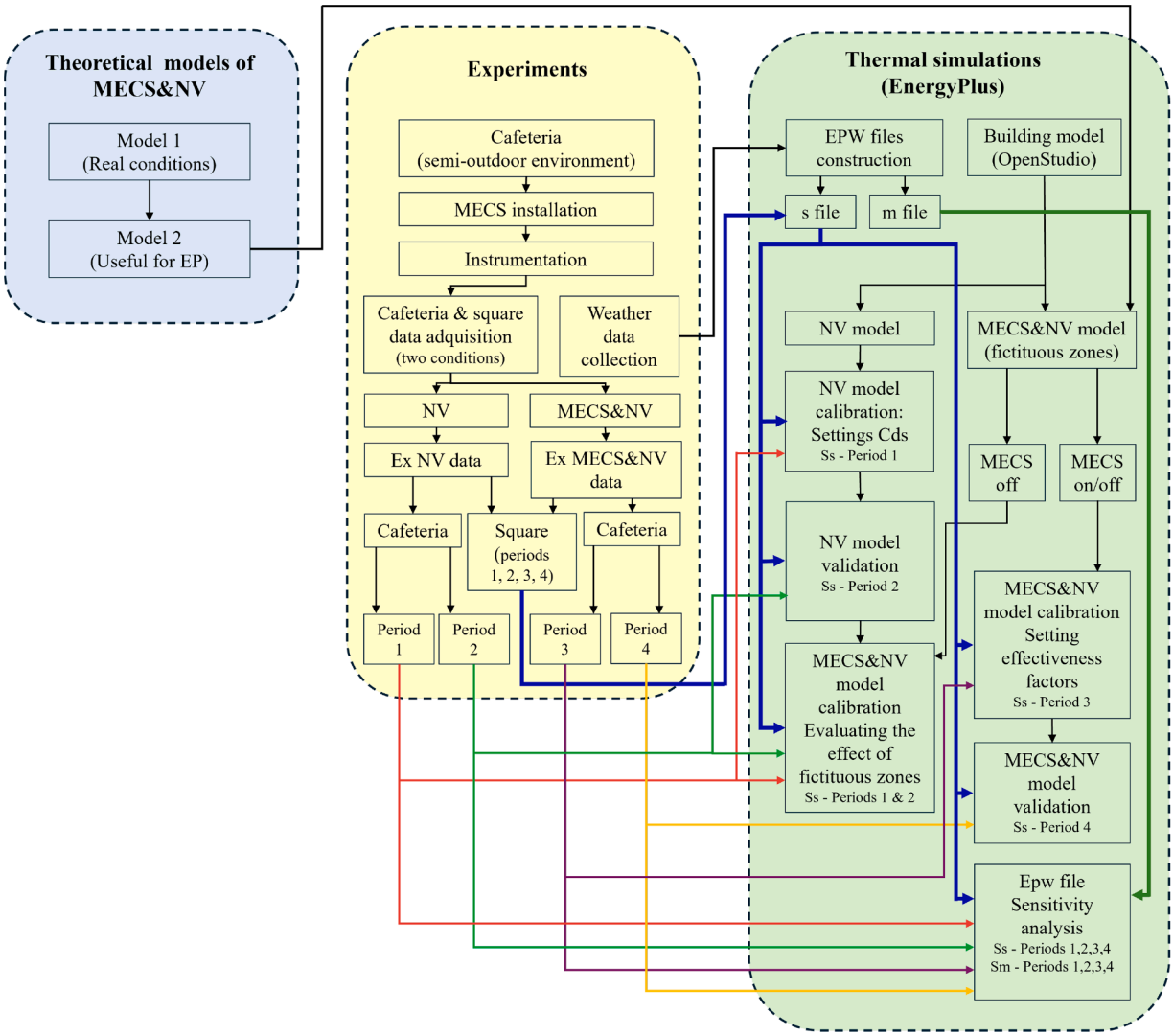


Fig. 1. Flow chart of the methodology. Acronyms: EnergyPlus (EP), EP weather (EPW) file, experimental (Ex), mist evaporative cooling system (MECS), natural ventilation (NV), simulation using s file (Ss), and simulation using m file (Sm).

affecting the performance of the system and the water flow to the nozzles.

3. Methodology

The methodology considered in this study includes the development of two theoretical models of MECS&NV; experiments performed in a semi-outdoor environment, where a MECS with two misting lines was implemented; and thermal simulations of this semi-outdoor environment. The semi-outdoor environment is the cafeteria of the Renewable Energy Institute of the National Autonomous University of Mexico (IER-UNAM). The flow chart of the methodology is presented in Fig. 1.

Theoretical Model 1 represents the actual thermodynamic process, which unfortunately could not be implemented in EnergyPlus (EP) for the reasons explained below. Model 2 mimics Model 1 results, with the advantage that it can be implemented in EP. Experiments with MECS in operation (MECS&NV) and with the MECS out of operation and only with natural ventilation (NV) were carried out in a semi-outdoor environment, the cafeteria. The experimental results were used for the calibration and validation of the thermal simulation models for the NV condition and for the MECS&NV condition.

3.1. Theoretical models of MECS&NV process

This section presents two theoretical models of the MECS&NV process. Model 1 represents the actual thermodynamic process; however, it cannot be implemented in EnergyPlus. Model 2 employs a fictitious system that replicates the results of Model 1 and can be implemented into EnergyPlus.

3.1.1. Model 1–Actual thermodynamic process

In Model 1, air enters a zone where water droplets from the nozzles evaporate; this zone is called the evaporative zone, the air reduces its temperature due to the evaporation processes and increases its relative humidity. Model 1 is illustrated in Fig. 2; this diagram corresponds to the actual configuration of each mist line of the MECS.

The variables at the inlet of the evaporative zone known from the experiments are the wind speed ($v_{a,i}$), air dry-bulb temperature ($T_{db,i}$), air relative humidity (ϕ_i), atmospheric pressure (P_{atm}), and the inlet opening area (A_i). Also from experiments, the water volumetric flow rate delivered by the nozzles ($\dot{V}_{w,i}$) is know.

The evaporative process is isenthalpic; consequently, the air enthalpy (h) remains constant and the wet-bulb temperature (T_{wb}) is

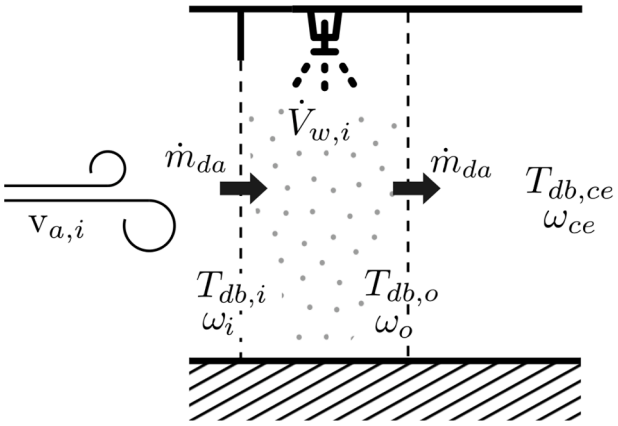


Fig. 2. Model 1 diagram. The area between the dashed lines represents the evaporative zone. Water droplets from the nozzles enter with a volumetric flow rate ($\dot{V}_{w,i}$). The inlet air to the evaporative zone has wind speed ($v_{a,i}$), dry-bulb temperature ($T_{db,i}$) and humidity ratio (ω_i). The outlet air exiting the evaporative zone and entering to the cafeteria has dry-bulb temperature ($T_{db,o}$) and humidity ratio (ω_o). The dry air mass flow rate ($\dot{m}_{da}$) entering and exiting the evaporative zone is the same. The cafeteria air has dry-bulb temperature ($T_{db,ce}$) and humidity ratio (ω_{ce}).

almost constant,

$$h = h_i = h_o \quad \text{and} \quad T_{wb} \simeq T_{wb,i} \simeq T_{wb,o}, \tag{1}$$

where h_i and $T_{wb,i}$ are the specific enthalpy and wet-bulb temperature of the inlet air, and h_o and $T_{wb,o}$ are the specific enthalpy and wet-bulb temperature of the outlet air. The inlet variables h_i and $T_{wb,i}$ can be calculated from the experimentally measured variables using psychrometric equations [36].

According to mass conservation, the dry air mass flow rate remains constant,

$$\dot{m}_{da} = \dot{m}_{da,i} = \dot{m}_{da,o}, \tag{2}$$

and the sum of the water vapor mass flow rate at the inlet air ($\dot{m}_{v,i}$) and the inlet water mass flow rate from the nozzles ($\dot{m}_{w,i}$) is equal to the water vapor mass flow rate at the outlet air ($\dot{m}_{v,o}$) plus the mass of the non-evaporated water flow rate from the nozzles ($\dot{m}_{w,o}$),

$$\dot{m}_{v,i} + \dot{m}_{w,i} = \dot{m}_{v,o} + \dot{m}_{w,o}. \tag{3}$$

The dry air mass flow rate can be calculated as

$$\dot{m}_{da} = \frac{\dot{V}_{a,i}}{\upsilon_{da,i}}, \tag{4}$$

where $\dot{V}_{a,i}$ is the inlet air volumetric flow rate and $\upsilon_{da,i}$ is the inlet air specific volume, which is calculated using psychrometric equations [36]; $\dot{V}_{a,i}$ can be obtained in terms of experimentally measured variables as

$$\dot{V}_{a,i} = C_d A_i v_{a,i}, \tag{5}$$

where C_d is the discharge coefficient of the inlet opening.

The inlet air water vapor mass flow rate is calculated as

$$\dot{m}_{v,i} = \dot{m}_{da}\, \omega_i, \tag{6}$$

where ω_i is the inlet air humidity ratio. It can be calculated from the measured psychrometric variables and psychrometric equations [36].

The inlet water mass flow rate through the nozzles is given by

$$\dot{m}_{w,i} = \rho_w\, \dot{V}_{w,i}, \tag{7}$$

where ρ_w is the density of water.

When inlet velocity is too high, part of the water droplets from the nozzles can leave the evaporative zone before evaporating, and when inlet velocity is too low, the droplets can evaporate completely within the zone, but part of the cooled air can exit through the same inlet, reducing the cooling capacity of the system. Therefore, from the conservation of water, Eq. (3), and assuming that all droplets remaining in the evaporative zone evaporate, the maximum water vapor mass flow rate at the outlet air ($\dot{m}^*_{v,o}$) can be expressed as

$$\dot{m}^*_{v,o} = \dot{m}_{v,i} + \epsilon_m \dot{m}_{w,i}, \tag{8}$$

where ϵ_m is the misting line effectiveness. Note that the superscript * is introduced to indicate that all droplets remaining in the evaporative zone does evaporate, *i.e.* the air is not saturated.

The maximum outlet air humidity ratio (ω^*_o) is calculated as

$$\omega^*_o = \frac{\dot{m}^*_{v,o}}{\dot{m}_{da}}. \tag{9}$$

The dry bulb temperature of the outlet air is calculated using the following psychrometric equation [36],

$$T_{db,o} = \frac{h - 2501\ \omega^*_o}{1.006 + 1.86\ \omega^*_o}. \tag{10}$$

When $T_{db,o} \geq T_{wb,i}$, the air is not saturated, therefore $\omega_o = \omega^*_o$, $\dot{m}_{v,o} = \dot{m}^*_{v,o}$, and $T_{db,o}$ is given by Eq. (10).

In contrast, when $T_{db,o} < T_{wb}$, the air is saturated, hence, $T_{db,o} = T_{wb}$, and ω_o and $\dot{m}_{v,o}$ are calculated as

$$\omega_o = \frac{h - 1.006\ T_{wb}}{2501 + 1.86\ T_{wb}} \tag{11}$$

and

$$\dot{m}_{v,o} = \omega_o \dot{m}_{da}. \tag{12}$$

Fig. 3 shows the Model 1 process in the psychrometric chart.

The evaporative cooling effectiveness of the MECS process in the cafeteria (ϵ_{ce}) is the ratio between the change in the cafeteria air dry bulb temperature obtained in the evaporative process and the maximum possible temperature change,

$$\epsilon_{ce} = \frac{T_{db,c} - T_{db,ce}}{T_{db,i} - T_{wb}}. \tag{13}$$

3.1.2. Model 2–Useful for implementation in EnergyPlus

Model 1 can not be implemented into EnergyPlus because EnergyPlus only allows air mass flow to enter to a thermal zone. Therefore, Model 2 is designed to replicate the air conditions at the exit of the evaporative zone given by Model 1, this model requires a fictitious system, an ideal evaporative cooling system (IECS) is used for this purpose. In Model 2, the input to the evaporative zone is not the water mass flow rate of the nozzles, but an air mass flow rate supplied by this IECS, as shown in Fig. 4.

The input to the IECS is outdoor air with dry-bulb temperature ($T_{db,i}$), humidity ratio (ω_i), dry air mass flow rate ($\dot{m}_{da,e}$), and water volumetric flow rate ($\dot{V}_{w,i}$). While the output from the IECS, that enters the evaporative zone is an air with a dry-bulb temperature ($T_{db,e}$), humidity ratio (ω_e), and with the same dry air mass flow rate ($\dot{m}_{da,e}$) which remains constant during the evaporative process. Air also enters directly from the outdoor to the evaporative zone with wind speed ($v_{a,i}$), dry-bulb temperature ($T_{db,i}$), and humidity ratio (ω_i). In the evaporative zone, the mixture of air from the IECS and air from the outdoor produces that the air exiting the evaporative zone has dry-bulb temperature ($T_{db,o}$), humidity ratio (ω_o) and dry air mass flow rate ($\dot{m}_{da,o}$). Due to the air exiting the evaporative zone that enters the cafeteria, the air in the cafeteria achieves a dry-bulb temperature ($T_{db,ce}$) which is lower than it would have been if the evaporative process had not occurred.

In Model 2, the IECS airflow outlet follows the same isoenthalpic line as in Model 1 (see Figs. 3 and 5); therefore, the mixture of this airflow with the actual inlet airflow into the evaporative zone is also aligned with this isoenthalpic line, and the relationship between the evaporative dry air mass flow rate ($\dot{m}_{da,e}$) and the inlet dry air mass flow rate ($\dot{m}_{da}$) is

$$\frac{\dot{m}_{da,e}}{\dot{m}_{da,o}} = \frac{T_{db,i} - T_{db,o}}{T_{db,o} - T_{db,e}} = \frac{\omega_i - \omega_o}{\omega_o - \omega_e}, \tag{14}$$

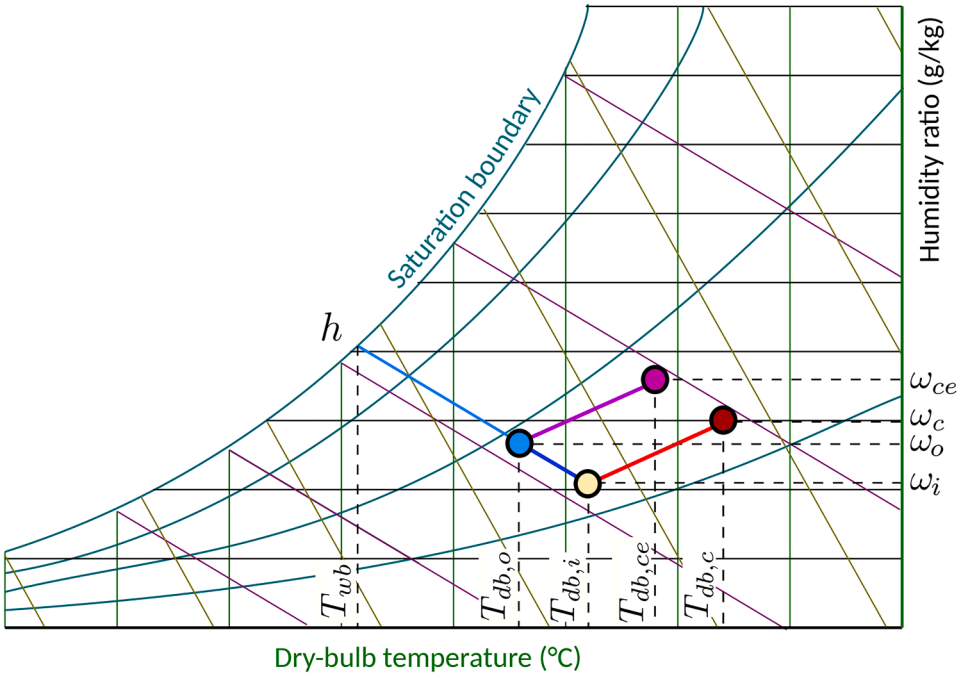


Fig. 3. Model 1 process in the psychrometric chart. The inlet air to the evaporative zone (yellow point) has dry-bulb temperature ($T_{db,i}$) and humidity ratio (ω_i). The evaporative process is isoenthalpic with specific enthalpy (h) (blue line). The outlet air of the evaporative zone (blue point) has has dry-bulb temperature ($T_{db,o}$) and humidity ratio (ω_o). The air exiting the evaporative zone enters the cafeteria, producing air with $T_{db,ce}$ and ω_{ce} (magenta point). If the air exiting the evaporative zone did not undergo the evaporative process, the air in the cafeteria would have $T_{db,c}$ and ω_c (dark red point). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Equipment and uncertainty associated with each variable measured in the cafeteria and the square.

Variable	Equipment	Uncertainty
Temperature (Tc)	Type T thermocouple	±0.2 °C (10–50 °C)
Humidity (Tc)	Fluke 975 AirMeter	±2% (10–90%)
Wind speed (Ac & As)	Gill Instruments WindMaster 3D Ultrasonic Anemometer	±1.5% RMS (at 12 m/s)
Wind direction (Ac & As)	Gill Instruments WindMaster 3D Ultrasonic Anemometer	±2° (at 12 m/s)

where $\dot{m}_{da,e}$ is calculated by

$$\dot{m}_{da,e} = \frac{\dot{V}_{a,i,e}}{v_{da,i}}. \tag{15}$$

The inlet air volumetric flow rate in the IECS ($\dot{V}_{a,i,e}$) is calculated assuming that it can be expressed as

$$\dot{V}_{a,i,e} = C_{ac} V, \tag{16}$$

where C_{ac} is the air changes per unit of time and V the volume of the evaporative zone. C_{ac} is obtained in a calibration process.

To ensure the same inputs to the evaporative zone as in Model 1, the water volumetric flow rate supplied to the IECS must be equal to the actual volumetric flow rate delivered by the nozzles ($\dot{V}_{w,i}$).

Since the IECS output replicates that of Model 1, the resulting equations are similar to Eqs. (6) and (8), (9), (10) (11). Therefore, the inlet vapor mass flow rate entering the IECS ($\dot{m}_{v,i,e}$) is expressed as

$$\dot{m}_{v,i,e} = \dot{m}_{da,e}\ \omega_i. \tag{17}$$

The maximum outlet water vapor mass flow rate in IECS ($\dot{m}^*_{v,o,e}$) is calculated as

$$\dot{m}^*_{v,o,e} = \dot{m}_{v,i,e} + \epsilon_I\ \dot{m}_{w,i}, \tag{18}$$

where $\dot{m}_{w,i}$ is calculated using Eq. (7), and the effectiveness of the IECS (ϵ_I) is obtained through a calibration process. It is a value to adjust the evaporative outlet air conditions of Model 2 to be the same of those obtained in Model 1. $T_{db,o}$ can be considered the same as that in Model 1 if internal heat gains in the evaporative zone are negligible.

The maximum humidity ratio at the outlet of IECS (ω^*_e), that is supplied to the evaporative zone, is

$$\omega^*_e = \frac{\dot{m}^*_{v,o,e}}{\dot{m}_{da,e}}. \tag{19}$$

The dry-bulb temperature at the outlet of IECS ($T_{db,e}$) is obtained as

$$T_{db,e} = \frac{h - 2501\ \omega^*_e}{1.006 + 1.86\ \omega^*_e} \tag{20}$$

If $T_{db,e} < T_{wb}$, it means that the air is saturated, then, $T_{db,e} = T_{wb}$ and the humidity ratio is calculated as

$$\omega_e = \frac{h - 1.006\ T_{wb}}{2501 + 1.86\ T_{wb}}. \tag{21}$$

The misting line effectiveness (ϵ_m) of each evaporative zone (ϵ_{ez}) is calculated with Eq. (8) and the evaporative cooling effectiveness of the MECS in the cafeteria (ϵ_{ce}) is calculated using Eq. (13).

Note that Model 2 implies that $\dot{m}_{da,e}$ is added to the evaporative zone. However, in EnergyPlus the air flow balance of the Airflow Network does not include the HVAC Zone equipment, therefore, from this balance $\dot{m}_{da,i} = \dot{m}_{da,o} = \dot{m}_{da}$.

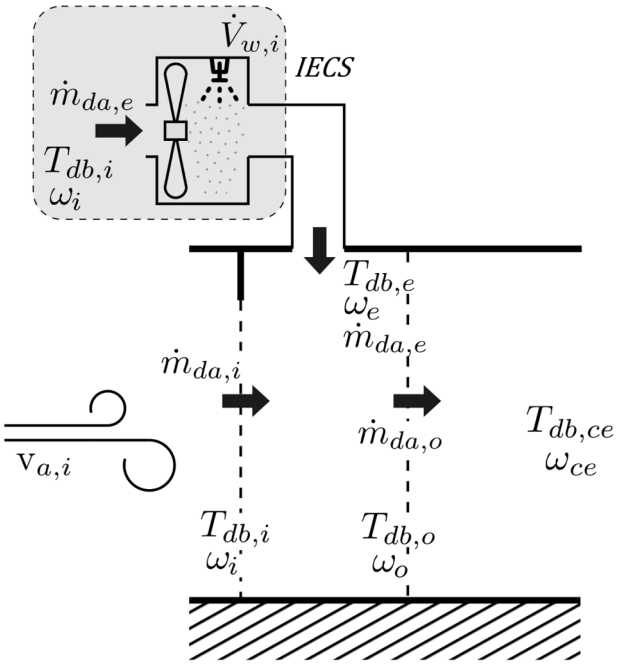


Fig. 4. Model 2 diagram. The area between the dashed lines represents the evaporative zone. An ideal evaporative cooling system (IECS) is added. The input of the IECS is air from the outdoor with dry-bulb temperature ($T_{db,i}$), humidity ratio (ω_i), dry air mass flow rate ($\dot{m}_{da,e}$) and water volumetric flow rate ($\dot{V}_{w,i}$). The output of the IECS that enters the evaporative zone is an airflow with dry-bulb temperature ($T_{db,e}$), humidity ratio (ω_e), and a dry air mass flow rate ($\dot{m}_{da,e}$). Inlet air to the evaporative zone has wind speed ($v_{a,i}$), dry-bulb temperature ($T_{db,i}$), humidity ratio (ω_i), and dry air mass flow rate ($\dot{m}_{da,i}$). The outlet air exiting the evaporative zone has dry-bulb temperature ($T_{db,o}$), humidity ratio (ω_o) and dry air flow mass rate ($\dot{m}_{da,o}$). The air in the cafeteria has dry-bulb temperature ($T_{db,ce}$) and the humidity ratio (ω_{ce}).

3.2. Experiments

The experiments were conducted in the cafeteria of the IER-UNAM at Temixco, Morelos, Mexico.

3.2.1. Cafeteria–Semi-outdoor environment

The cafeteria is a semi-outdoor environment because it has no wall at the entrance facing a square, and on the opposite side, it has a high porosity lattice wall. It is part of a four-story building in sloping terrain. The cafeteria is located on the first floor. There are two stories above the cafeteria and one semi-buried story below.

The cafeteria has a floor area of 126 m^2 and a height from floor to the ceiling of 3.36 m. It has a reticular steel beam system that gives an effective height for the air flow of 2.90 m. The main facade, the entrance, faces east with an angle of 6.8° to the south and is completely open. The rear facade has a clay brick lattice wall with a porosity of 0.38, open to airflow. On the north side is the kitchen, separated from the cafeteria by a half-height wall and a steel curtain; on the south side, the facade is a wall made of a hollow clay brick wall. Fig. 6 shows the floor plan and a cross-section of the cafeteria. In the experiments reported here, neither the cafeteria nor the kitchen was occupied, there were no internal heat gains, and the steel curtain remained closed.

3.2.2. MECS Installation

The MECS installed in the cafeteria of the IER-UNAM has two misting lines placed 0.33 m from the ceiling, each line with 14 nozzles 0.50 m apart from each other. One line was placed at 0.70 m from the entrance to the cafeteria and the other at the rear, at 0.35 m from the lattice wall (see Fig. 6). A Fogco misting system pressurized the water at 1000 PSI; 144 L/h of water was supplied to the misting lines. The nozzles produce droplets approximately 5 μm in diameter.

3.2.3. Instrumentation

The air temperature in the cafeteria was measured employing a 2.90 m high tower with six thermocouples. It was placed near the cafeteria center (point Tc) in Fig. 6. The first thermocouple was at the top, and the rest were 0.30 m apart on a vertical line. Temperature data were recorded through a data acquisition system with open hardware and software. Relative humidity was measured at the center of the cafeteria (point Tc) with a Fluke meter with its acquisition system. In addition, a three-component ultrasound anemometer was used, with its center at a height of 1.45 m, the middle effective height of the cafeteria (point Ac) in Fig. 6. To measure outdoor air velocity, a similar anemometer was placed in the square (point As), also shown in Fig. 6, aligned with the first and 8.85 m from the cafeteria entrance. These anemometers were connected to computers for data acquisition. The velocity of the outdoor air on the side of the lattice wall was impossible to measure due to the sloping terrain. Table 1 summarizes the equipment used and the associated measurement uncertainties for each experimental variable. Photographs of the instrumentation and the misting lines in operation are shown in Fig. 7.

3.2.4. Cafeteria and square data acquisition

Measurements were carried out over three days in February 2025 (15, 16, and 17) during the dry-hot season, with neither occupancy nor other internal heat gains. Two conditions were considered, with the MECS out of operation, that is, with only natural ventilation (NV), and with the MECS in operation (MECS&NV). An experimental data file was constructed for each one of these two conditions. Air velocity data measured in the square were used to build one of the two weather files (Section 3.3.1). For each condition, the experimental data from the cafeteria were divided into two periods, one used for the calibration of the model and the other for its validation.

3.2.5. Weather data collection

The weather data for the entire February 2025 were taken from a meteorological station located in the upper part of a building next to the cafeteria building. Fig. 8 shows an aerial photograph of the IER-UNAM, where the cafeteria building and the meteorological station building are indicated. Data measured in the meteorological station were used to build the two weather files (Section 3.3.1).

3.3. Thermal simulations

The thermal simulations of the cafeteria were performed using EnergyPlus 22.2 (EP). The conduction transfer function method was used to compute heat flux through the building surface constructions. The TARP and DOE-2 algorithms were used for the convection in the indoor and outdoor surfaces, respectively. The timestep between calculations was one minute.

3.3.1. Weather files construction

Two EnergyPlus weather (EPW) files were constructed for the thermal simulations in EP. The first EPW file is constructed employing all data from the meteorological station located in the upper part of a building next to the cafeteria building. This EPW file is named the *m* file.

Although the meteorological station is very close, due to all the buildings and trees around (see Fig. 8), the wind speed and wind direction measured at this meteorological station are expected to be different from those reaching the cafeteria entrance or the lattice wall. Therefore, an other EPW file has been constructed, in which the data of the outdoor air velocity measured in the square, were used to substitute the wind speed and direction, this EPW file is named the *s* file. The sensitivity of the simulation results to the use of these two EPW files is reported in Section 4.9.

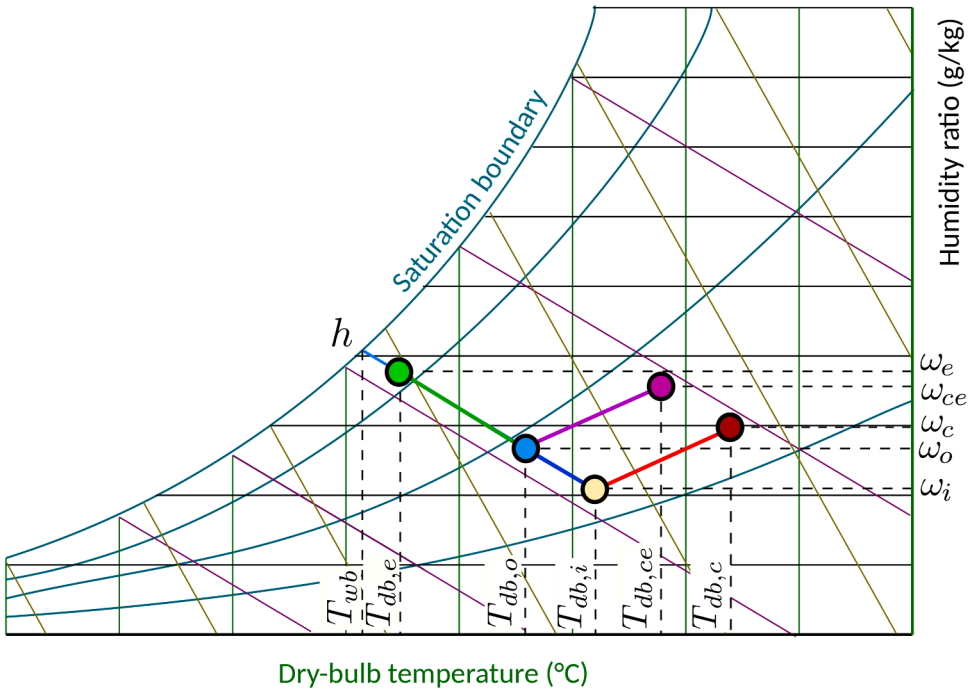


Fig. 5. Model 2 process in the psychrometric chart. The inlet air to the evaporative zone (yellow point) has dry-bulb temperature $T_{db,e}$ and humidity ratio ω_e. The evaporative process is isoenthalpic with enthalpy h (blue line). The outlet air from the IECS that enters the evaporative zone has $T_{db,o}$ and ω_o (point green). The outlet air of the evaporative zone (blue point) has $T_{db,o}$ and ω_o. The air exiting the evaporative zone enters the cafeteria and produces the air in the cafeteria has $T_{db,ce}$ and ω_{ce} (magenta point). If the air exiting the evaporative zone did not undergo the evaporative process, the air in the cafeteria would have a higher dry temperature $T_{db,c}$ and a lower humidity ratio ω_c (dark red point). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.3.2. Building model

The model of the cafeteria building was created using the OpenStudio SketchUp plug-in, it is shown in Fig. 9.

In the building model, the ground is considered at the level of the semi-buried story floor, and the square is considered as the roof of a thermal zone with exterior boundary conditions. The lattice wall, on the west, was simulated with an equivalent shading system. In the square, there is a tree that has been simulated with two perpendicular shading surfaces, as suggested in [37].

The construction of the cafeteria walls is composed of a 14 cm layer of brick (thermal conductivity $k = 0.7$ W/m K, density $\rho = 1970$ kg/m^3, and specific heat $C_p = 800$ J/kg K) and the construction of the floor and the ceiling is made of a 16 cm concrete layer ($k = 2.0$ W/m K, $\rho = 2400$ kg/m^3, and $C_p = 1000$ J/kg K). The wall that separates the cafeteria from the kitchen has in the upper part a metal curtain that was closed in the experiments ($k = 45.2$ W/m K, $\rho = 7824$ kg/m^3, and $C_p = 500$ J/kg K).

3.3.3. NV Model

To simulate natural ventilation in the proposed NV model, the EnergyPlus AirFlow Network (AFN) model was used, applying the cross-ventilation model for air movement in the cafeteria. The pressure coefficients required in the AFN model are taken from the ones calculated by EP [28].

The cross-ventilation algorithm is capable to calculate the air velocity of the jet when the air velocity in the thermal zone is large enough to apply the model.

The discharge coefficient (C_d) of the entrance and that of the lattice wall were set through a calibration process (Section 4.2).

3.3.4. MECS&NV Model

The scheme of the MECS&NV model implemented in EnergyPlus is presented in Fig. 10. One of its key features is the creation of a thermal zone representing the evaporative zone associated with each MECS misting line. In the case of the cafeteria, one thermal zone corresponds to the entrance misting line and the other to the lattice wall misting line. Therefore, the total cafeteria volume is divided into three thermal zones: the Lattice evaporative zone, the Cafeteria zone, and the Entrance evaporative zone. The openings between the Evaporative zones and the Cafeteria zone are fully open windows, and for radiation calculation a glazing system with solar heat gain coefficient of 0.999 and a brick frame of 0.05 m, which adds a negligible thermal mass, where considered.

As mentioned in Section 3.1.2, an Ideal Evaporative Cooling System (IECS) is required to emulate the behavior of the misting system. Therefore, the object ZoneHVAC:ForcedAir:UserDefined is used to incorporate the IECS, as it allows the implementation of a custom Energy Management System (EMS) program, for each one of the evaporative thermal zones. This EMS program is used to define the conditions of the supply air node in the evaporative zone. The AFN model was used to simulate natural ventilation. For the AFN model, five nodes are considered: two outdoor nodes, one at the entrance and another at the lattice wall, and three interior nodes, one in each of the evaporative zones and another in the cafeteria zone. The airflow balance in the evaporative zone is performed without considering the mass airflow from the ZoneHVAC entering through the supply air node. The pressure coefficients used for the outdoor nodes are calculated by EnergyPlus.

The EMS program employs a plug-in programmed in Python, named Mist Evaporative Cooling Plug-in. It assigns the conditions of the supply air node of the evaporative zone based on outdoor conditions and the water flow rate of the misting line. Fig. 11 shows the flowchart of the Mist Evaporative Cooling Plug-In and its interaction with EnergyPlus. EP activates the plug-in at each timestep after EP has completed its initial calculations. Then, EP starts the function on_user_defined_component_model. A conditional clause asks: Are there handles? If it is false (the first timestep), a set of handles is assigned to read the following variables from EnergyPlus: S, $T_{db,i}$, T_{wb}, ϕ_i, and p_{atm}.

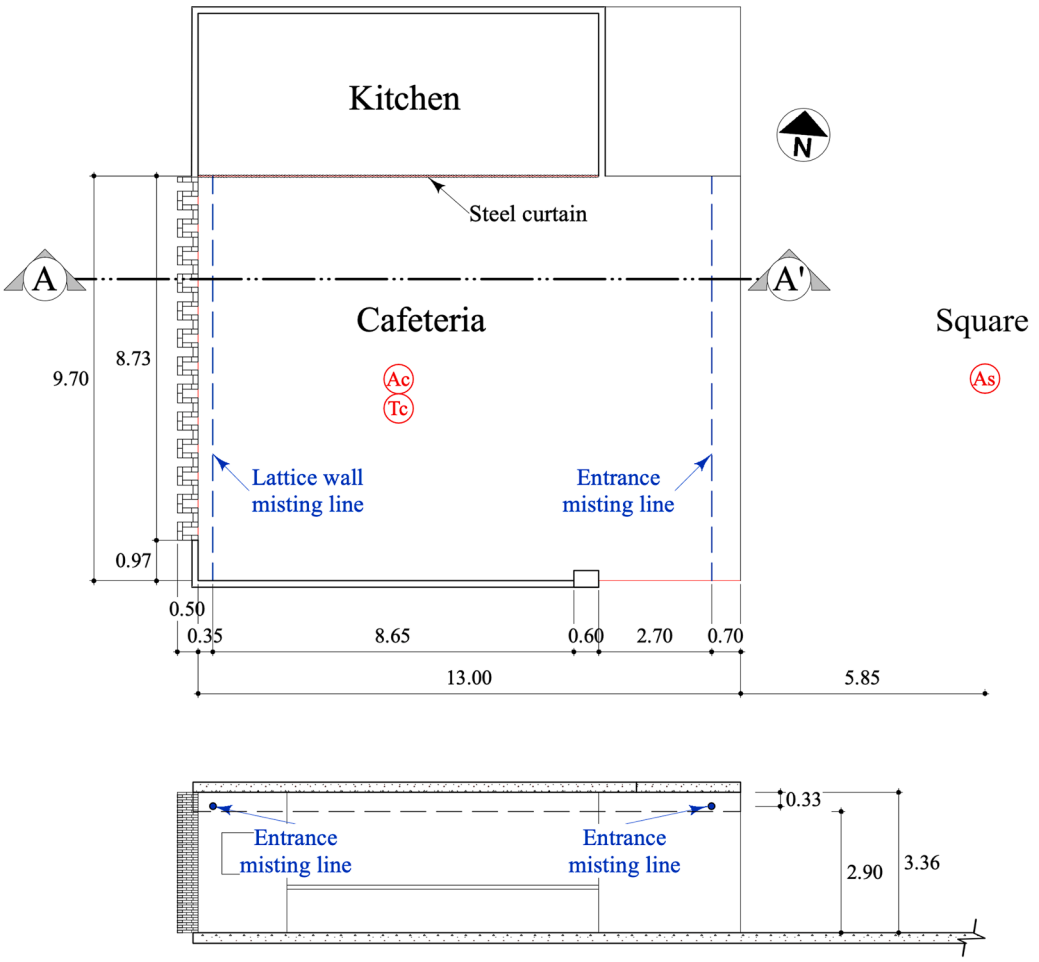


Fig. 6. Floor plan (up) and A-A' section plan (down) of the IER-UNAM cafeteria. The two misting lines, the position of the tower with thermocouples (Tc), and the two anemometers (Ac and As) are indicated.

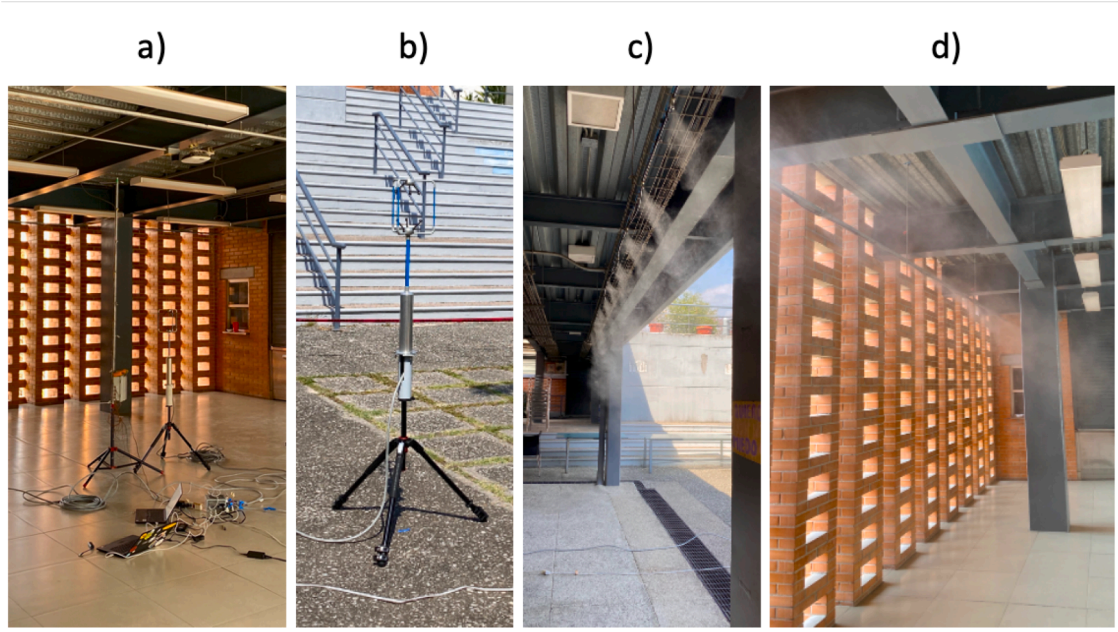


Fig. 7. Photographs of the experiments. a) Cafeteria instrumentation, b) anemometer at the square, c) entrance misting line, and d) lattice wall misting line.



Fig. 8. Aereal photograph of the IER-UNAM, the cafeteria building (C) and meteorological station building (M) are indicated. Photograph property of IER-UNAM.

S is a variable to indicate the misting line state (0 for off and 1 for on) assigned in a schedule. Then it leads to obtain the values of these variables using the handles. Otherwise, if it is true (as it is in the subsequent runs), it leads directly to obtaining these variable values using the handles. A conditional clause checks the S value: if $S = 1$ is false, then the supply air node conditions are $T_{db,e} = T_{db_i}$, $\omega_e = \omega_i$, $\dot{m}_{da,e} = 0$. If it is true, the plug-in runs a subprogram, named Evaporative Zone Supply Air, to calculate $T_{db,e}$, ω_e, and $\dot{m}_{da,e}$. For both cases, the values are stored in EnergyPlus database ESO and then, the plug-in assigns the values to the supply air node of the evaporative zone.

The Evaporative Zone Supply Air subprogram called by the plug-in was developed in Python to calculate the $T_{db,e}$, ω_e, and $\dot{m}_{da,e}$ when the misting lines operate in any of the evaporative zones. These quantities are calculated using the psychrometric theory presented in Section 3.1. Fig. 12 shows the flowchart of this subprogram and its interaction with EnergyPlus. The subprogram reads the outdoor air variables from EnergyPlus: $T_{db,i}$, T_{wb}, and p_{atm}. Then, using EnergyPlus psychrometric library functions humidity_ratio_d and specific_volume, ω_i and $v_{a,i}$ are calculated, respectively. Next, the subprogram reads $\dot{V}_{a,i,e}$, an user-defined variable calculated using Eq. (16). Then $\dot{m}_{da,e}$ is calculated using Eq. (15) and $\dot{m}_{v,i,e}$ using Eq. (17). Next, $\dot{V}_{w,i}$ and ϵ_I, user-defined variables are read, the first comes from the experiments and the second from the calibration process. Then, the following variables are subsequently calculated: $\dot{m}_{w,i}$ (Eq. (7)), $\dot{m}^*_{v,o,e}$ (Eq. (18)), ω^*_e (Eq. (19)), h (Eq. (1)), and $T_{db,e}$ (Eq. (20)). Then, a conditional clause asks if $T_{db,e} > T_{wb}$. If it is true, the mist does not saturate the air; therefore, the value of $T_{db,e}$ is valid, $\dot{m}_{v,o,e} = \dot{m}^*_{v,o,e}$, and $\omega_e = \omega^*_e$. When it is false, the air is saturated; therefore, $T_{db,o} = T_{wb}$, and ω_e is calculated using Eq. (21).

3.3.5. MECS&NV Model calibration process

The model calibration consists of two steps. The first step is to evaluate the effect of dividing the cafeteria into three thermal zones on the air volumetric flow rate within the cafeteria. The second step considers a value for the air changes per unit time in the evaporative zone (C_{ac}) and adjusts the IECS effectiveness (ϵ_I).

As was pointed out, dividing the cafeteria into three thermal zones in the EP model implies considering two new openings. To ensure that the airflow conditions inside the cafeteria are not affected by these openings, simulations using the MECS&NV model with the MECS turned off and simulations using the NV model were compared.

In the second calibration step, a value of C_{ac} is assumed. Then, for a period with the MECS turned on, simulations are performed by varying the value of ϵ_I. The simulated air temperature inside the cafeteria is compared with the experimental measurements, and the selected value of ϵ_I is the one that produces the closest agreement with the experimental results.

4. Results

The first three subsections present the results for the NV condition: the NV experimental results, the calibration of the NV model to determine the discharge coefficient values, and the validation of the calibrated NV model. The following five subsections correspond to the results for the MECS&NV condition: the calibration of the MECS&NV model to evaluate the effect of the evaporative zones as thermal zones, the experimental results, the calibration to determine the effectiveness factor values, the validation of the MECS&NV model, and the cooling effect of the MECS in the air inside the cafeteria. For all these analyses, the EPW s file is used (see Section 3.3.1). The final subsection presents a sensitivity analysis of the simulation results to the EPW file.

4.1. NV Experimental results

The entrance-perpendicular-component of the air velocity measured in the cafeteria, $V_{\perp c-Ex}$, and at the square, $V_{\perp o-Ex}$, as functions of time during the NV experiments are shown in Fig. 13, period 1 and period 2 are presented. Positive values indicate that the airflow is from the square. In general, when $0 < V_{\perp o-Ex}$, $V_{\perp o-Ex} < V_{\perp c-Ex}$, indicating that when the airflow enters the cafeteria through the entrance there is an increase of the air velocity in the cafeteria, which could be produced by the reduction of the entrance area compared with the outdoor area. While, when $V_{\perp o-Ex} < 0$, that is, when the airflow is from the lattice wall side, $|V_{\perp c-Ex}| < |V_{\perp o-Ex}|$ showing, as expected, that the lattice wall is an obstacle for the airflow.

The air temperature measured at the center of the cafeteria T_{c-Ex} is presented in Fig. 14, alongside the outdoor air temperature, recorded at the meteorological station T_{o-Ex}. Analysis of the data reveals that T_{c-Ex} is similar to T_{o-Ex}, indicating a well-ventilated space. The difference, taking T_{o-Ex} as reference, has MAE = 0.5 °C and MBE = 0.4 °C. In this figure, the relative humidity measured at the center of the cafeteria, ϕ_c, and the outdoor relative humidity, $\phi_{o\text{-}Ex}$, are also shown. These curves reflect the corresponding variations caused by temperature changes when the humidity ratio remains constant, as it is when there is no precipitation. It can be observed that during solar hours, the air inside the cafeteria exhibits slightly higher relative humidity than the outdoor air, with differences of up to 9%.

4.2. NV Model calibration: setting discharge coefficients

Simulations in EnergyPlus employing the cross ventilation algorithm provide the air velocity of the jet, V_j, only when the air velocity in the thermal zone is large enough; in the study case, this does not always occur, and therefore, this variable cannot be used to calibrate the NV model. Thus, calibration is performed by comparing only the simulation and experimental results of the cafeteria air temperature. The simulations consider T_{c-Ss}, and the experiments consider T_{c-Ex} (measured at point Tc). The EPW s file and Period 1 data are used for this calibration process.

The discharge coefficient (C_d) of the air entering through the entrance, where obstructions are small, was varied in the range of 0.50 to 0.60, while C_d of the air entering through the lattice wall ranged from 0.30 to 0.45. The set of values that has the minimum mean absolute error (MAE) and mean bias error (MBE) is $C_d = 0.55$ for the entrance and $C_d = 0.35$ for the lattice wall (MAE = 0.3 °C and MBE = 0.0 °C). Therefore, these discharge coefficient values are used in the subsequent simulations.

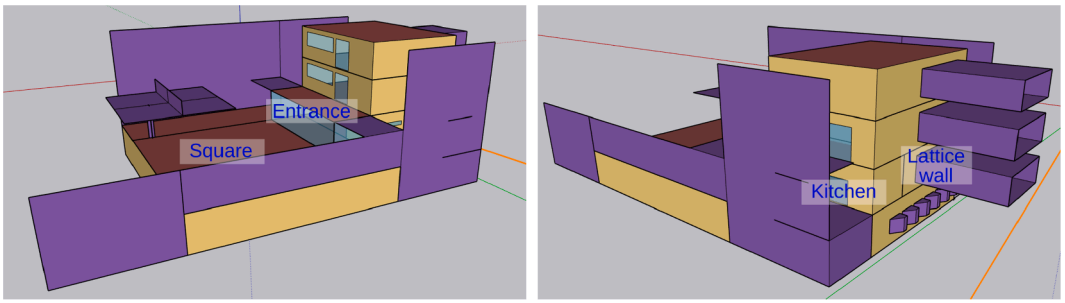


Fig. 9. Building model in OpenStudio. Neighboring buildings are represented as shading surfaces. (left) View of the main facade; the cafeteria is located at the square level, where a tree is also represented as shading surfaces. (right) View of the rear facade; the cafeteria is on the second floor. The shading surface systems at the cafeteria and the two upper levels provide an equivalent shading effect to that of the lattice walls. Shading surfaces are in purple. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

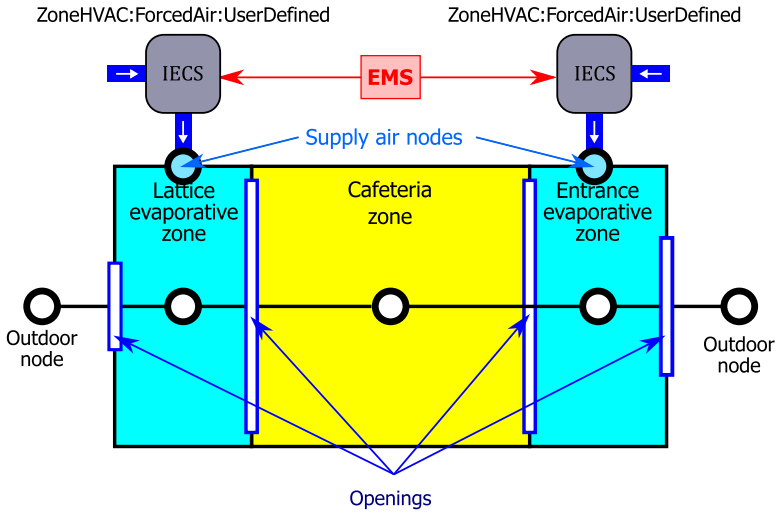


Fig. 10. Scheme of the MECS&NV model in EnergyPlus for the cafeteria. The total cafeteria volume is divided into three thermal zones: the Lattice evaporative zone, the Cafeteria zone, and the Entrance evaporative zone. For each evaporative zone, an Ideal Evaporative Cooling Systems (IECS) is used. The ZoneHVAC:ForcedAir:UserDefined object incorporates the IECS. An Energy Managment System (EMS) program calculates the supply air node conditions. For natural ventilation simulation, the Airflow Network (AFN) model has five nodes and four openings.

4.3. NV Model validation

For the validation of the NV model, the cafeteria air temperature and relative humidity results obtained in the simulations using the EPW *s* file during Period 2 are compared with the experimental values measured in the cafeteria center.

Temperature and relative humidity results of Period 2 are also shown in Fig. 14. It can be seen that there is good qualitative agreement between the simulation and the experimental results. To facilitate the comparison between simulation and experimental results, scatter plots of temperature and relative humidity are presented in Fig. 15. These plots show that the simulated temperature does not exhibit a systematic deviation from the experimental values, whereas the simulated relative humidity consistently underestimates the experimental measurements in about 4%. The differences between simulation and experimental results for temperature and relative humidity, and the entrance-perpendicular component of the air velocity in the cafeteria, as functions of time for Period 2, are presented in Fig. 16. It can be observed that the simulation provides better agreement (i.e., smaller differences in both temperature and relative humidity) when the entrance-perpendicular component of the air velocity in the cafeteria has relatively large values with small variations (Feb 17 from 2:00 to 7:00 hours).

The quantitative comparison of simulation results with the experimental ones for the temperature gives MAE = 0.2 °C and MBE = 0.1 °C and for the relative humidity MAE = 4% and MBE = -4%. The NV model is validated based on these results.

4.4. MECS&NV Model calibration: evaluating the effect of three thermal zones

To analyze whether there is any effect of considering the evaporative zone as a thermal zone on the air volumetric flow rate, $\dot{V}$, simulations are performed for Periods 1 and 2, with the NV model and with the MECS&NV model with the MECS off. As the separation of the evaporative zone and the cafeteria zone is fictitious, the discharge coefficient for the opening between the evaporative zone and the cafeteria zone is considered 1.0 for both evaporative zones. EP reports the air volumetric flow rate entering and exiting the cafeteria through the entrance, $\dot{V}_{e-i}$ and $\dot{V}_{e-o}$ respectively, and those entering and exiting through the lattice wall, $\dot{V}_{w-i}$ and $\dot{V}_{w-o}$. Taking as reference the results of the NV model (without evaporative zones), the MAE and MBE normalized with the mean value of the air volumetric flow rate entering to the cafeteria is smaller than 5%. This fact demonstrates that the effect of dividing the cafeteria into three thermal zones can be considered negligible.

4.5. MECS&NV Experimental results

The entrance-perpendicular component of the air velocity measured in the cafeteria during the MECS&NV experiments, $V_{\perp c\text{-}Ex}$, and that

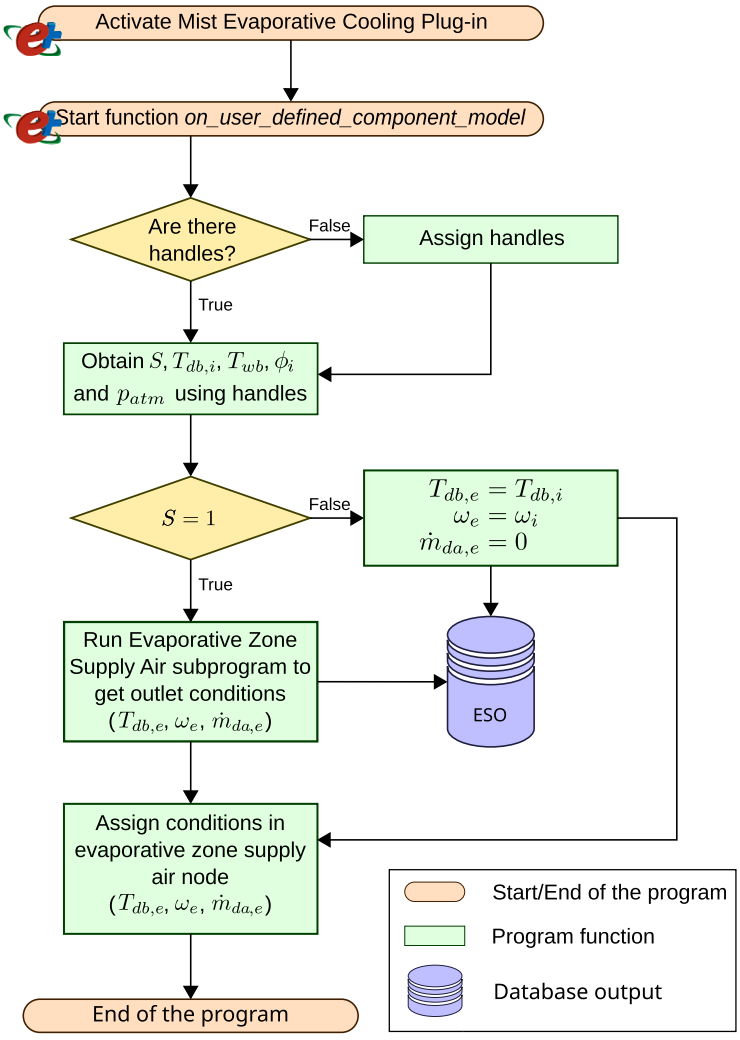


Fig. 11. Flowchart of the mist evaporative cooling plug-in and its interaction with EnergyPlus, denoted with the logo.

measured in the square, $V_{\perp o-Ex}$, as functions of time, are shown in Fig. 17. As expected, the relationship between these variables is similar to that observed in the NV experiments.

In the MECS&NV experiments, the air temperature T_{c-Ex} and relative humidity ϕ_{c-Ex} within the cafeteria were also monitored over time, the results are presented in Fig. 18, alongside the outdoor values T_{o-Ex} and ϕ_{o-Ex}. There are two periods during which the MECS operated; these periods are denoted in light green. Before the MECS begin to operate, T_{c-Ex} is close to T_{o-Ex}. While, ϕ_{c-Ex} is slightly higher than ϕ_{o-Ex}, the difference is up to 7%, a value similar to that obtained in the NV experiments (P1 and P2). When MECS begin to function, T_{c-Ex} decreases and ϕ_{c-Ex} increases. However, the temperature decrease and the humidity increase are not maintained equally when MECS are operating. The cause of this inconstant effect of the MECS is the air velocity, shown in Fig. 17. It can be seen that the effect of MECS is more pronounced when $|V_{\perp c-Ex}|$ is low, and the direction is from the cafeteria entrance.

4.6. MECS&NV Model calibration: setting the effectiveness

Before setting the IECS effectiveness (ϵ_I), the value of the air changes per unit time in the evaporative zone (C_{ac}) has to be assigned. It is considered that in a timestep (1 minute) the entire volume of the evaporative zone is changed, *i.e* $C_{ac} = 1$ timestep^{-1}). The selection of ϵ_I value is performed by comparing the simulation result of the cafeteria air temperature T_{c-Ss} with the experimental value T_{c-Ex}. For this calibration process, the EPW *s* file is used and data from Period 3. Since the characteristics of both misting lines are similar, ϵ_I was considered equal for both evaporative zones and was varied in the range of 0.30 to 0.60. The value that presents the minimum MAE and MBE is $\epsilon_I = 0.40$ (MAE = 0.2 °C and MBE = 0.1 °C). Therefore, this effectiveness value is used in subsequent simulations.

4.7. MECS&NV Model validation

For the validation of the MECS&NV model, the cafeteria air temperature and relative humidity results obtained in the simulations using the EPW *s* file during Period 4 are compared with the experimental values measured in the cafeteria center, data shown in Fig. 18. It can be seen that there is good qualitative agreement between the simulation and the experimental results. As in Section 4.3, to help in the comparison between simulation and experimental results, scatter plots of temperature and relative humidity are presented in Fig. 20. The ranges of the variables in these plots are the same as in Fig. 15, but the results are within a shorter range, as these data correspond only to the hottest period of the day, when the use of the MECS is adequate. As for the NV model validation, the simulated temperature does not exhibit a systematic deviation from the experimental values, and the simulated relative humidity underestimates the experimental measurements. Nevertheless, for the NV&MECS model, the underestimation is not constant, it increases as the relative humidity increases. The differences between

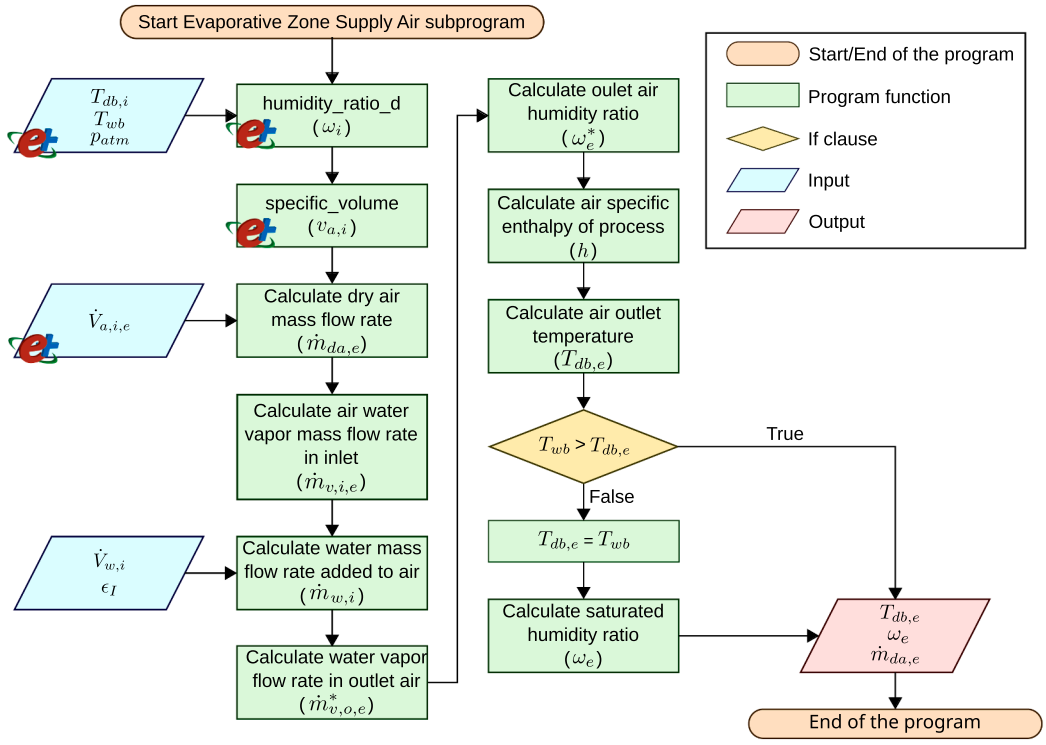


Fig. 12. Flowchart of the Evaporative Zone Supply Air subprogram, in Python, and its interaction with EnergyPlus.

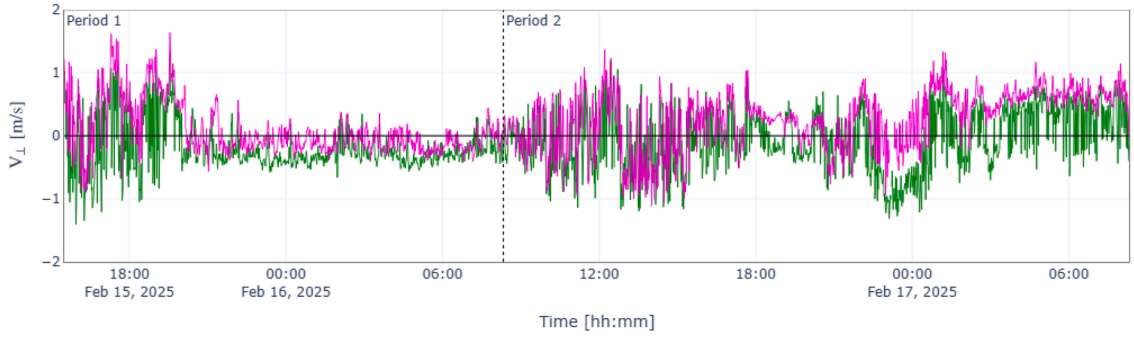


Fig. 13. NV periods. The entrance-perpendicular-component of the air velocity measured in the cafeteria, $V_{\perp c-Ex}$ (magenta), and at the square, $V_{\perp o-Ex}$ (green), as functions of time. Positive values indicate that the airflow is from the square. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

simulation and experimental results for temperature and relative humidity, and the entrance-perpendicular component of the air velocity in the cafeteria, as functions of time for Period 4, are presented in Fig. 21. It can be observed that the largest differences in temperature and relative humidity (1.5 °C and -18%) are obtained at the beginning of the operation of the MECS and when the entrance-perpendicular component of the air velocity in the cafeteria has relatively small values. This relationship between simulation and experimental differences and air velocity in the MECS&NV period is similar to that found in the NV period reported in Section 4.3. The quantitative comparison of simulation results with the experimental ones for the temperature gives MAE = 0.4 °C and MBE = 0.1 °C and for the relative humidity MAE = 5% and MBE = -4%. It is also concluded that the MECS&NV model is validated.

4.8. Cooling effect of the MECS in the cafeteria air

From the experimental results, the cooling effect of the MECS on the air inside the cafeteria can be evaluated by comparing the reduction of the cafeteria air temperature $T_{c\text{-}Ex}$ with respect to outdoor air

Table 2
Sensitivity to EPW file: *s* and *m* files. Metrics of comparison of temperature (T) and relative humidity (ϕ) simulation results with experimental ones, for the four periods. Experimental results of the cafeteria air velocity are included: the prevailing direction of the flow entering to the cafeteria (θ_p): from the entrance E or from the lattice wall W, the percentage of time (t_p) and the average of the perpendicular velocity ($\bar{v} = \overline{V_{\perp c-Ex}}$) when being in that direction.

Period	Metric	T [∘C]		ϕ [%]		θ_p [-]	t_p [%]	$\bar{v}$ [m/s]
		s	*m*	*s*	*m*			
P1	MAE	0.5	0.3	4	4	W	55	-0.19
	MBE	0.5	0.2	−4	−4			
P2	MAE	0.4	0.3	5	4	E	73	0.51
	MBE	0.4	0.2	−5	−4			
P3	MAE	0.2	0.5	4	5	E	70	0.74
	MBE	0.1	0.5	−3	−4			
P4	MAE	0.4	0.6	5	6	E	70	0.80
	MBE	0.1	0.6	−4	−6			

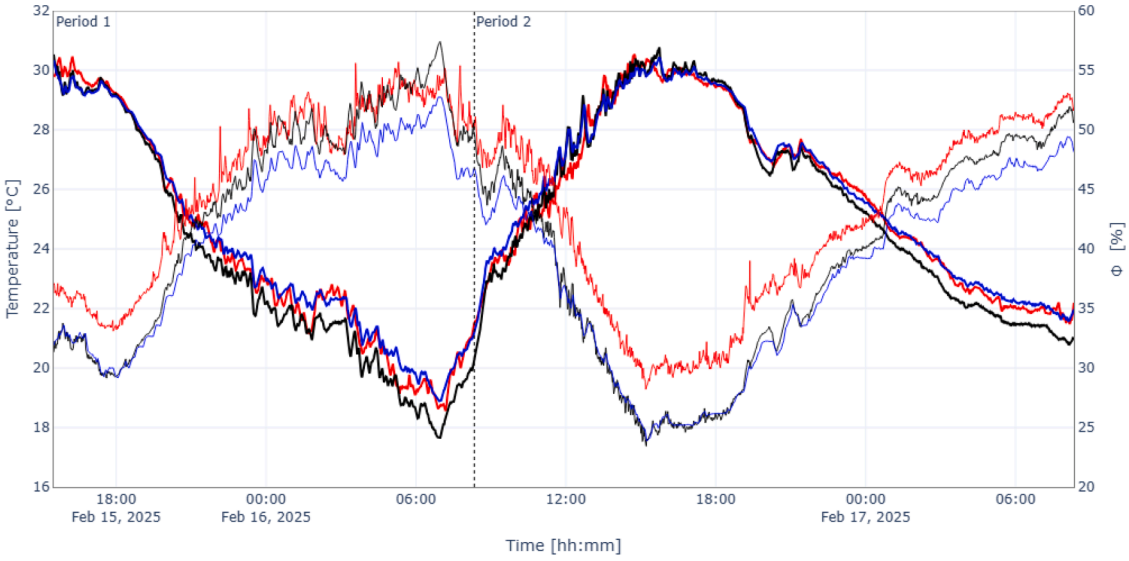


Fig. 14. NV periods. Air temperature T (thick lines) and relative humidity ϕ (thin lines) results. From measurements in the cafeteria center T_{c-Ex} and ϕ_{c-Ex} (red), in the outdoor, T_{o-Ex} and ϕ_{o-Ex} (black); and from simulations using EPW s file T_{c-Ss} and ϕ_{c-Ss} (blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

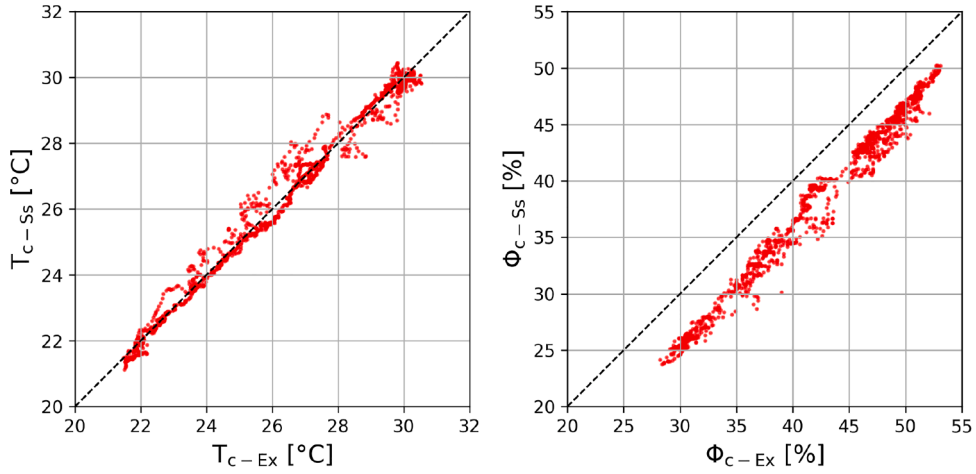


Fig. 15. NV Period 2. Scatter plots showing the relationship between experimental and simulated results, for cafeteria air temperature (left) and relative humidity (right).

temperature $T_{o\text{-}Ex}$. A maximum temperature reduction of 3.1 °C is observed, accompanied by a maximum increase of $\phi_{c\text{-}Ex}$ with respect to $\phi_{o\text{-}Ex}$ of 24%. During MECS operation, the average difference between the cafeteria temperature and the outdoor temperature is 1.9 °C and the average difference between $\phi_{c\text{-}Ex}$ and $\phi_{o\text{-}Ex}$ is 11%. After the MECS is turned off, the indoor temperature remains more than 1 °C below the outdoor temperature for 6 minutes in Period 3 and for 12 minutes in Period 4.

From the simulations, the cooling effect of the MECS, also relative to the outdoor conditions, results in a maximum air temperature reduction of 3.0 °C with a corresponding maximum increase in relative humidity of 11%. The average air temperature reduction is 1.9 °C, with an average relative humidity increase of 7%. These results are in agreement with the experimental ones.

The cooling effect attributable solely to the MECS, obtained through simulations, can be evaluated by comparing the results obtained with the MECS operating to those of the same periods with the MECS turned off. The maximum air temperature reduction is 2.9 °C, with a corresponding maximum increase in relative humidity of 11%. The average air temperature reduction is 1.9 °C, and the average relative humidity increase is 7%. These values are similar to those obtained when referenced to outdoor conditions, indicating that the thermal mass of the cafeteria has a minor contribution to the air temperature reduction within the space compared to the evaporative cooling effect.

The average misting line effectiveness (ϵ_m) (Eq. (8)) for the semi-outdoor cafeteria, during periods P3 and P4 when the lines are active, is 0.18 for the entrance misting line and 0.17 for the lattice misting line. This type of space is difficult to analyze since a large proportion of the misting water droplets do not evaporate inside the cafeteria. The selected location of the misting lines, near the cafeteria entrance and near the lattice wall, may be one of the causes of the reduced effectiveness of the water used. It was also observed that the contribution of the misting lines to the cafeteria's cooling was affected by the wind direction and airflow turbulence. For the same periods P3 and P4, the average evaporative cooling effectiveness of the MECS in the cafeteria (ϵ_{ce}) (Eq. (13)) is 0.16. This value is largely a consequence of the low effectiveness of the misting line ϵ_m. Therefore, the cooling effect of the MECS in the cafeteria could be improved by relocating the misting lines in an intermediate distance between the center and the entrance or the lattice wall.

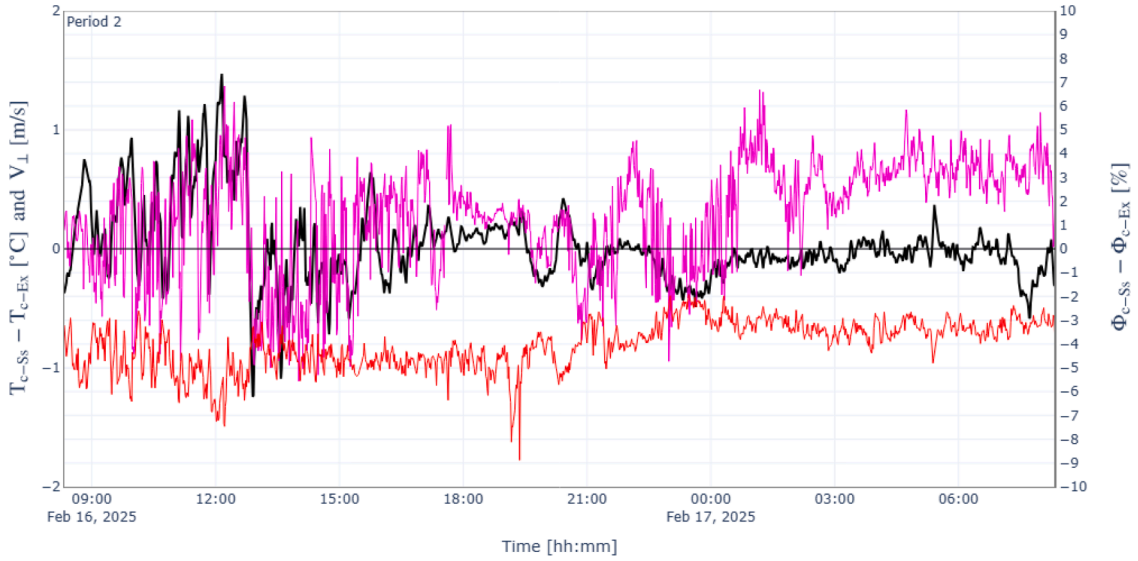


Fig. 16. NV Period 2. Differences between simulation and experimental results for temperature, $T_{Ss} - T_{Ex}$ (black), and relative humidity, $\phi_{Ss} - \phi_{Ex}$ (red), and the entrance-perpendicular component of the air velocity in the cafeteria $V_{\perp c-Ex}$ (magenta), as functions of time. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

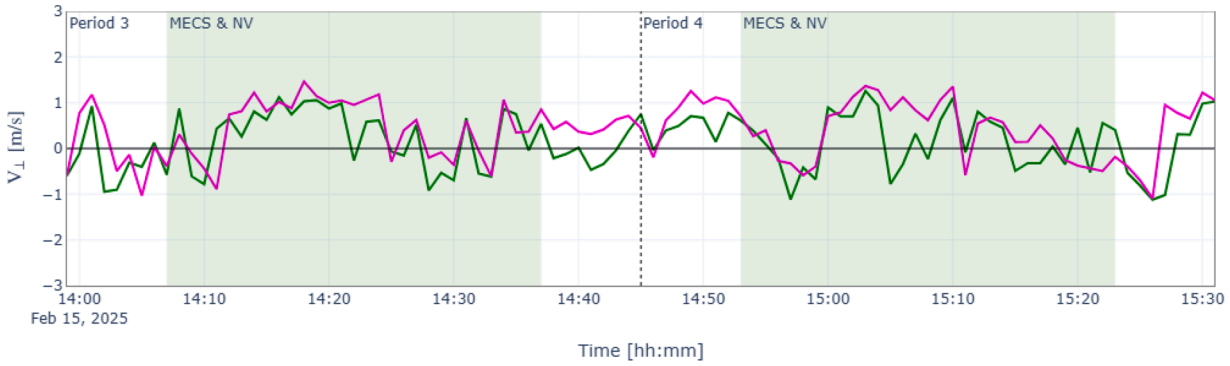


Fig. 17. MECS&NV periods. The entrance-perpendicular-component of the air velocity measured in the cafeteria, $V_{\perp c-Ex}$ (magenta), and at the square, $V_{\perp o-Ex}$ (green), as functions of time. Positive values indicate that the airflow is from the square. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.9. EPW File sensitivity analysis

As stated before, two EPW files were constructed (Section 3.3.1), but in previous simulations, only the EPW *s* file was used.

Since the outdoor air velocity is the only input data that differs in both EPW files, the entrance-perpendicular-component of the incident air velocity to the cafeteria as a function of time from both EPW files are compared. In the Ss simulations, $V_{\perp o-Ex}$ was measured in the square and the corresponding velocity in the Sm simulation, $V_{\perp Smh}$, was obtained from the meteorological station with the correction to have it at the same height of $V_{\perp o-Ex}$, both are shown for the four experimental periods in Fig. 19. It can be seen that there are differences between these velocities, as $V_{\perp Smh}$ does not account for the influence of neighboring buildings that alter the direction of the flow near the cafeteria building. When airflow is from the square, *i.e.* $0 < V_{\perp o-Ex}$, almost all the time $V_{\perp o-Ex} < V_{\perp Smh}$; and that in some periods when $V_{\perp Smh}$ has a small positive value, $V_{\perp o-Ex}$ is small but negative.

The effect of using the EPW *m* file on the MECS&NV model simulations is analyzed by comparing simulation results with the experimental ones using MAE and MBE metrics for the air temperature and relative humidity in the cafeteria, which are presented in Table 2. All simulations overestimate T and underestimate ϕ. It can be seen that, results from the *s* and *m* files have similar MAE and MBE values. The maximum difference for the air temperature in MAE is 0.3 °C and in MBE 0.5 °C, and for the relative humidity in MAE it is 1% and in MBE 2%. This indicates that the sensitivity to the EPW file choice is small, despite the complicated geometry of the nearby buildings surrounding the cafeteria. As expected, for P1 the simulation results are more accurate for the *m* file, when the air in the cafeteria comes from the lattice wall; and for P3 and P4 the simulation results are more accurate for the *s* file when the air comes from the square. Nevertheless, for P2 the *m* file produces slightly more accurate results, even though the air also comes from the square; the differences remain small, up to 0.2 °C for T and 1% for ϕ.

5. Discussion and limitations

The cooling effect in the air of the cafeteria produced by the MECS, of approximately 2 to 3 °C, is a good result compared to the results of previous studies taking into account that the cafeteria is a semi-outdoor space with a volume of 545 m^3. The high cooling effect reported by Pearlmutter et al. [16], of up to 15 °C, corresponds to the cooling of the air within the cooling tower, not to the air inside the people-occupied space. In the study by Rabani et al. [17], they report a cooling effect in the air inside a test room of up to 8 °C, produced by a water spray system inside a Trombe wall in combination with a solar chimney. Nevertheless, the test room has a small volume of less than 17 m^3. The cooling effect of

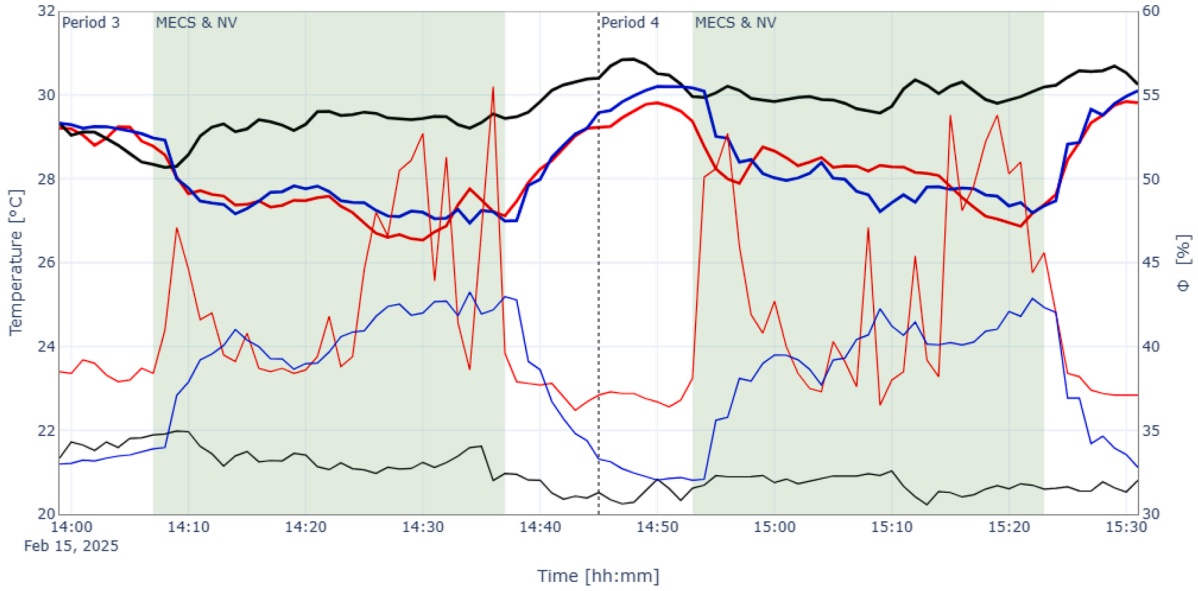


Fig. 18. MECS&NV periods. Air temperature T (thick lines) and relative humidity ϕ (thin lines) results. From measurements in the cafeteria center T_{c-Ex} and ϕ_{c-Ex} (red), in the outdoor, T_{o-Ex} and ϕ_{o-Ex} (black); and from simulations using EPW s file T_{c-Ss} and ϕ_{c-Ss} (blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

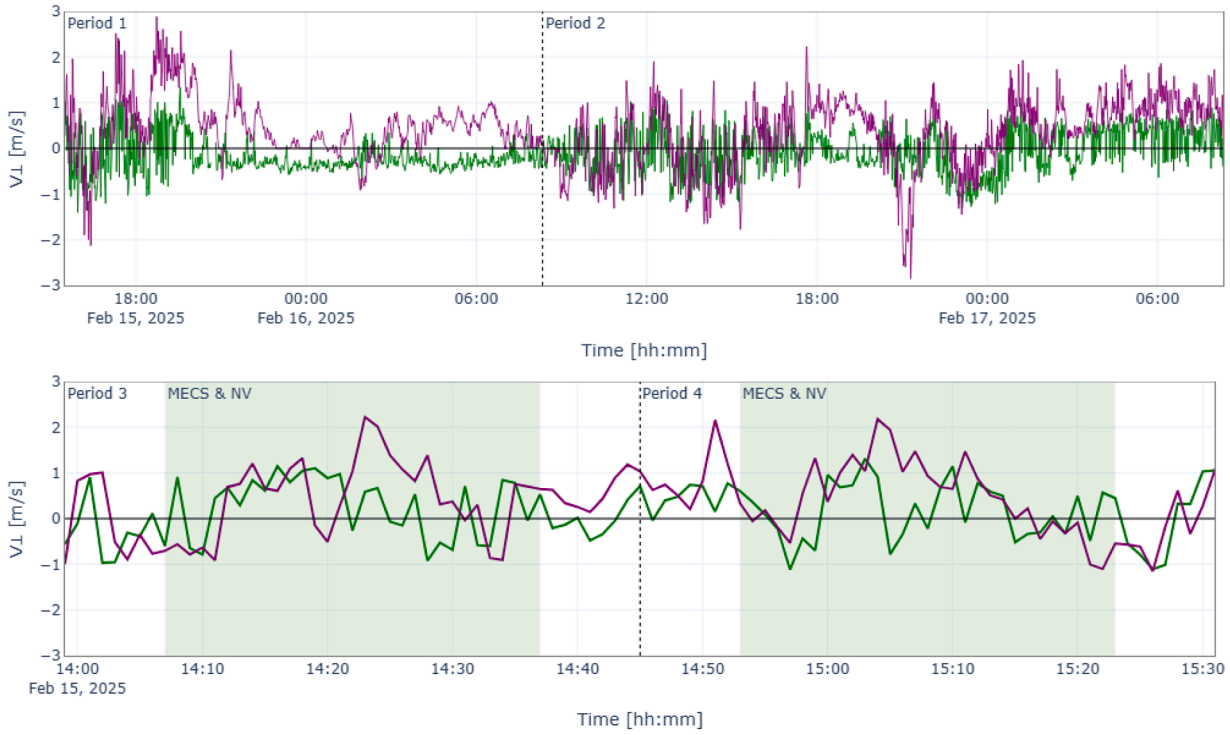


Fig. 19. NV and MECS&NV periods. The entrance-perpendicular-component of the incident air velocity to the cafeteria as a function of time in the experiments, which was used in the simulations Ss, $V_{\perp o-Ex}$ (green), and that in the simulation Sm, $V_{\perp Smh}$ (purple). Positive values indicate that the airflow is from the square. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

8 °C produced by an overhead mist cooling system in an outdoor space reported in Ulpiani et al. [21] corresponds to the effect in the air just down the mist. The cooling effect of an outdoor windcacher with a mist cooling system on an urban canyon reported by Kachmar et al. [23] is of 2 °C in the air temperature on the sidewalk close to the system and of 1 °C in the sidewalk far from the system.

The MECS being analyzed exhibits characteristics intrinsic to a system installed in a semi-outdoor environment with significant natural ventilation and no forced airflow. A particular characteristic that represents an additional challenge for the MECS is that the cafeteria is a space with large variations in air velocity due to the complex wind flow around it.

The investigation reported in this paper has some limitations. One of these is that EnergyPlus simulations assume perfect mixing of the air within each thermal zone, including the cafeteria, which has a large volume. Therefore, the indoor air temperature and relative humidity values

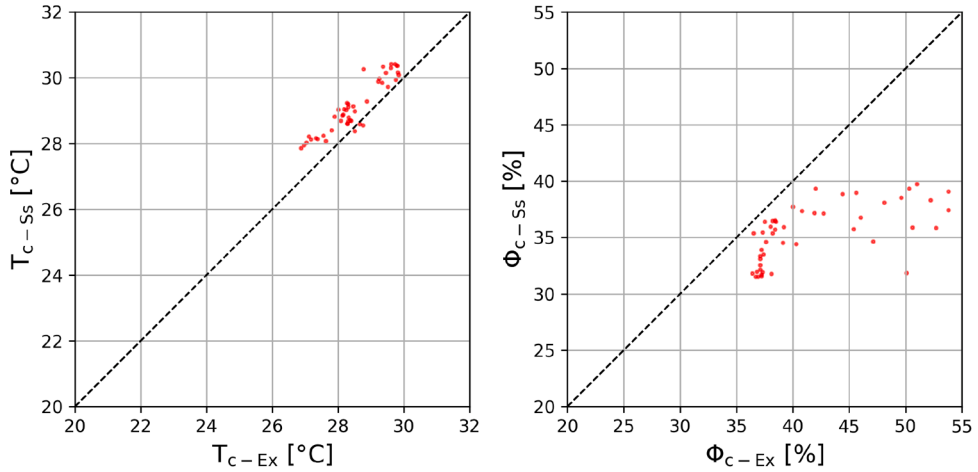


Fig. 20. MECS&NV Period 4. Scatter plots showing the relationship between experimental and simulated results in Period 4, for cafeteria air temperature (left) and relative humidity (right).

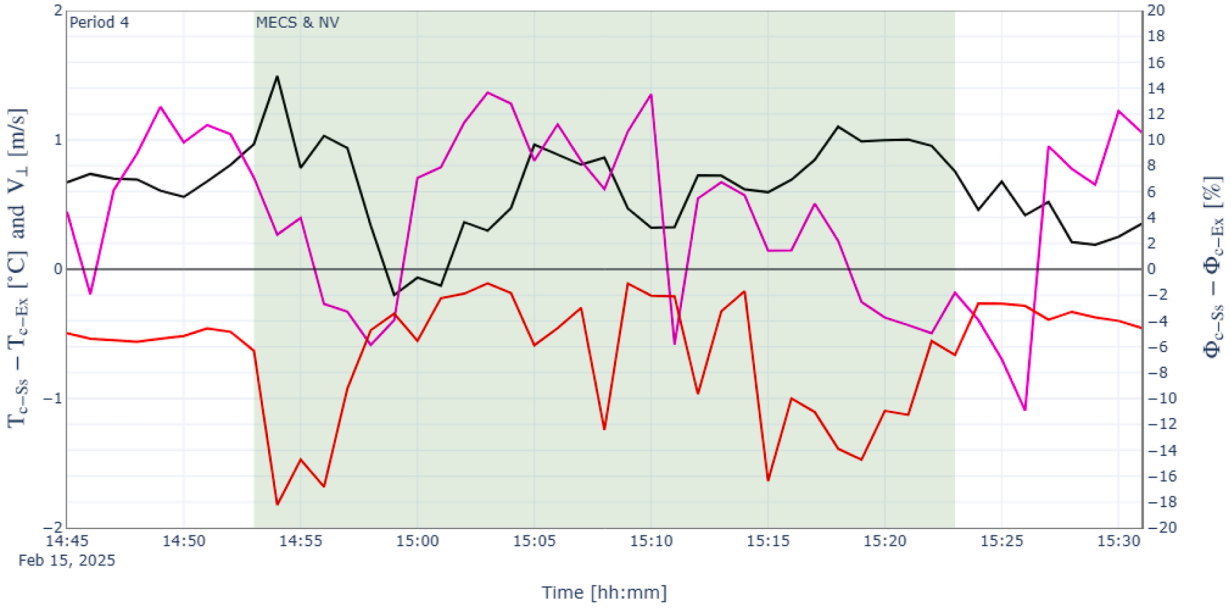


Fig. 21. MECS&NV Period 4. Differences between simulation and experimental results for temperature, $T_{Ss} - T_{Ex}$ (black), and relative humidity, $\phi_{Ss} - \phi_{Ex}$ (red), and the entrance-perpendicular component of the air velocity in the cafeteria $V_{\perp c-Ex}$ (magenta), as functions of time. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

obtained from the simulations correspond to the entire volume, whereas the measurements are point values in the center of the cafeteria, considered the most representative of the space. Consequently, EnergyPlus could not accurately capture some spatial and temporal variations in air temperature and relative humidity, which could limit the evaluation of the thermal comfort provided by the MECS in the cafeteria.

In this research, only periods of 30 minutes with mist-on and 15 minutes with mist-off were considered. Additional research is required to determine the optimal duration of these periods. Continuous mist-on operation would increase water consumption; however, it is not expected to saturate the air, as the cafeteria is a semi-outdoor space with significant airflow.

The present research has not evaluated changes in occupants' predicted thermal comfort or their perceived comfort. This is an important topic to be investigated in future work, as an increase in relative humidity can reduce the positive effect of the temperature decrease.

6. Conclusions

The main contribution of this research is the development of a model for the thermodynamic process of direct evaporative cooling produced by misting lines in a semi-outdoor space, implemented in EnergyPlus simulations. The model is validated through experiments conducted in a cafeteria. Experiments and simulations were performed under two conditions: natural ventilation only (NV), and operation of the mist evaporative cooling system combined with natural ventilation (MECS&NV).

The main conclusions are as follows.

- For the NV condition in the cafeteria, the air temperature from simulation with respect to that from experiments shows MAE = 0.2 °C and MBE = 0.1 °C.
- In the simulations, dividing the cafeteria into three thermal zones in the MECS&NV model results in changes of less than 5% in the

normalized air volumetric flow rate entering the cafeteria, relative to the NV model.

- For the MECS&NV condition in the cafeteria, the air temperature from simulations with respect to that from experiments shows MAE = 0.4 °C and MBE = 0.1 °C.
- The cooling effect of the MECS on the cafeteria air temperature compared with the value it would have without the MECS has a maximum reduction of 2.9 °C and an average of 1.9 °C.
- A sensitivity analysis was conducted comparing an EPW *m* file, containing data entirely from a meteorological station, with an EPW *s* file that incorporates air velocity measured in the square near the cafeteria entrance. The resulting differences in MAE and MBE with respect to experimental data are below 0.5 °C for temperature and below 2% for relative humidity.
- The cooling effect of the MECS in the cafeteria could be improved by relocating the misting lines in an intermediate distance between the center and the entrance or the lattice wall.
- The results of MECS simulations can be improved if a meteorological file with local wind data is available; however, the use of meteorological data from a nearby weather station can be used for decision-making regarding the sizing and implementation of this type of evaporative cooling system without significantly affecting the results.

It is worth noting that this model can be implemented in other whole-building simulation programs, provided that they offer the capability to control an Energy Management System (EMS) or similar from an external program.

CRediT authorship contribution statement

M.V. Cruz-Salas: Writing – review & editing, Visualization, Software, Formal analysis, Data curation; **E. Puerto:** Writing – review & editing, Writing – original draft, Software, Methodology, Conceptualization; **J. Rojas:** Writing – review & editing, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization; **G. Barrios:** Writing – review & editing, Methodology, Investigation, Data curation; **G. Huelsz:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization; **G. Ramírez:** Writing – review & editing, Investigation.

Data availability

We have shared the link to code and files at the Attach File step.

Declaration of competing interest

None.

Acknowledgments

Authors thank the technical support given by Guillermo Hernández and Kevin Alquicira, and Jesús Quiñonez for the meteorological data. This research was partially supported by PAPIIT-UNAM IT102622 project. M.V. Cruz-Salas acknowledges the postdoctoral fellowship granted by the SECIHTI.

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.enbuild.2026.117700.

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